THE GORENSTEIN PROPERTY OF A MORPHISM OF SPACES IS STABLE UNDER ARBITRARY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6669-6685 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 6687-6702, the complete author proof (14 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Paired-restatement check: the schemes chapter carries the same-named lemma; the conclusions differ in category (spaces vs schemes) and the spaces proof's chart-descent steps are not present schemes-side, so per the lane rule the two are counted separately with this difference noted. The schemes side is queued on its own merits.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definition of Gorenstein morphism of spaces (flat lfp with Gorenstein fibres, smooth-local on source and target), etale charts of algebraic spaces, and the schemes-side stability of the Gorenstein property under base change (cited from the schemes chapter).

Source reference. Original lemma 0C07: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 6702–6702. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume the fibres of f are locally Noetherian. Let $Y' \to Y$ be locally of finite type. Let $f' : X' \to Y'$ be the base change of f. Let $x' \in |X'|$ be a point with image $x \in |X|$.*

1. *If f is Gorenstein at x, then $f' : X' \to Y'$ is Gorenstein at x'.*
2. *If f is flat at x and f' is Gorenstein at x', then f is Gorenstein at x.*
3. *If $Y' \to Y$ is flat at $f'(x')$ and f' is Gorenstein at x', then f is Gorenstein at x.*

Proof. Denote $y \in |Y|$ and $y' \in |Y'|$ the image of x'. Choose a surjective étale morphism $V \to Y$ where V is a scheme. Choose a surjective étale morphism $U \to X \times_Y V$ where U is a scheme. Choose a surjectiev étale morphism $V' \to Y' \times_Y V$ where V' is a scheme. Then $U' = U \times_V V'$ is a scheme which comes equipped with a surjective étale morphism $U' \to X'$. Choose $u' \in U'$ mapping to x'. Denote

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$u \in U$ the image of u'. Then the lemma follows from the lemma for $U \to V$ and its base change $U' \to V'$ and the points u' and u (this follows from the definitions). Thus the lemma follows from the case of schemes, see Duality for Schemes, Lemma 0C07. $\square$